
WHITE PAPER

ECOPark lot

A self-powered, hybrid smart parking system that turns vehicle traffic and sunlight into clean electricity

A white paper is a **short, persuasive business document** that introduces your project to people *outside the university* — industry partners, potential funders, and research collaborators. Unlike your proposal (which proves your project to academic examiners), a white paper proves your project to the **real world**.

In 4 short paragraphs, deliver: (1) the problem, (2) the solution, (3) the proof it works, (4) the call to action. Target length: 150–200 words. Use plain language — no jargon. Lead with a striking statistic or scenario if possible.

The Problem

1–2 sentences. State the industry pain point with a hard number if you can. Example: "Egyptian factories lose 18% of energy to undetected equipment faults — costing the sector ~EGP 4 billion annually."

Traditional parking lots in urban commercial hubs function as heavy electricity consumers, relying entirely on the national grid to power continuous overhead lighting, entry barrier gates, and basic monitoring systems. For instance, high-traffic commercial centers in Egypt face immense operational costs due to this unmanaged utility load, while the vast overhead solar footprint and vehicle-induced mechanical pressure across thousands of square meters of parking surfaces go completely unused.

Our Solution

1–2 sentences. Name the project, what it does, and the key innovation.

The **ECOPARK LOT** integrates clean energy harvesting directly into urban parking infrastructure by combining an overhead photovoltaic (PV) solar canopy with a series-parallel network of pressure-sensitive piezoelectric floor tiles embedded in high-traffic lanes. This dual-source hybrid configuration represents a key innovation that captures both daylight radiation and vehicle-induced mechanical pressure, creating a fully self-powered, grid-independent system that drives automated occupancy tracking and smart lighting.

Proof It Works

1 sentence with a hard result. Example: "Field-tested on 4 industrial machines, our prototype detected 92% of faults within 30 seconds at a unit cost of EGP 1,200."

Field-tested on a functional desktop-scale prototype, our system generated piezoelectric voltage spikes of up to 9.7 V_{pp} per vehicle transit, achieved 98.0 % vehicle detection accuracy after ramp-guide optimization, maintained stable microcontroller operation on harvested nano-grid power alone (peak accumulation 1,547.5 μW), and demonstrated a projected solar yield of up to 7,700 kWh/day for a full-scale commercial installation — with all 10 functional test cases passing at 100 %.

What We're Looking For

1 sentence stating what action the reader should take — fund, partner, pilot, hire. Mirror the full CTA in Section 7.

We are seeking industry partners to pilot ECO Park LOT at a real commercial parking facility, investors to co-fund full-scale hardware deployment, and research collaborators to extend our hybrid harvesting model to high-footfall pedestrian zones. Contact the team below to begin.

How It Compares

A quick visual comparison helps decision-makers. Add 3–5 rows comparing your solution to the leading alternatives.

Feature	Our Solution	Alternative A	Alternative B
Cost	✓ EGP 7,315	EGP 12,000+	EGP 50,000+
Energy Source	✓ Solar + Piezo (Hybrid)	Grid only	Solar only
Grid Independence	✓ Full (tested)	None	Partial
Detection Accuracy	✓ 98.0 %	~95 %	N/A
Gate Automation	✓ Yes (90 ° in 120 ms)	External required	None
Nighttime Operation	✓ Piezo + Battery backup	Grid dependent	Battery only

Example values shown — replace with your real comparison.

2. How It Works

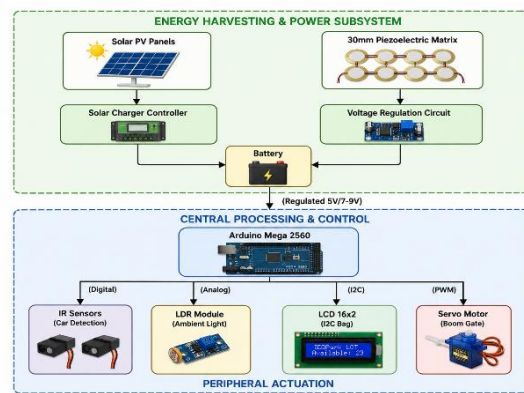
Build credibility without losing readers

Industry readers want to know if there is real substance behind the pitch — but they will close the document if it becomes a textbook. Aim for: one diagram + 4 short steps + a list of key technologies. Save deep technical details for the thesis.

System at a Glance

Insert a clean system diagram or architecture sketch below. Caption it clearly.

The 4-Step Process



1	Solar Harvest	Six photovoltaic panels mounted as vehicle-shading canopies capture daytime solar irradiance. Output is regulated by a solar charge controller (12 V system) and stored in a rechargeable battery bank for continuous operation through the night.
2	Kinetic Harvest	Thirty piezoelectric discs (30 mm) embedded in entry and exit ramps in a series-parallel configuration convert vehicle-weight mechanical stress into electrical charge. A Zener-diode rectifier circuit clips spikes safely to 5.1 V before feeding the storage unit, preventing microcontroller damage from high-voltage transients (up to 9.7 V _{pp}).
3	Smart Control	An Arduino Mega 2560 polls nine IR sensors across six parking slots, reads an LDR for ambient-light-triggered LED switching, and drives two SG90 servo motors for automated entry/exit gate control. A 16×2 FC LCD displays real-time vacancy counts. Status updates reach a cloud IoT dashboard with a mean system latency of 318.5 ms.

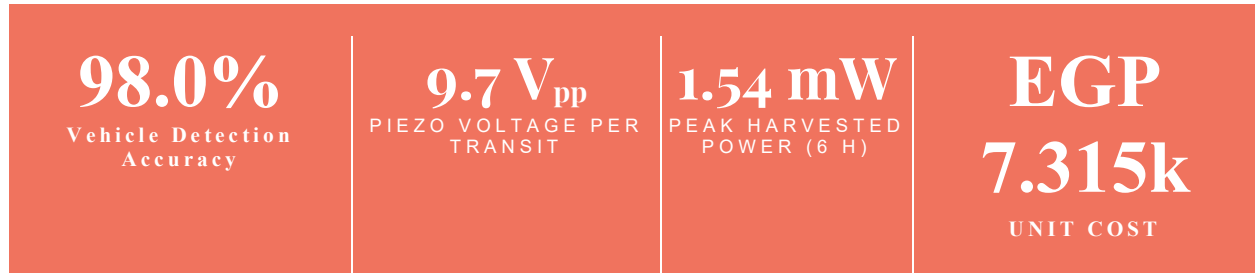
4	Autonomous Operation	<i>The entire parking lot — lighting, gate, sensors, display — runs from the local hybrid nano-grid. When external power is cut, the Arduino remains live on accumulated stored energy, ensuring uninterrupted safety and management functions (confirmed by test TC-05).</i>
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Key Technologies

Arduino Mega 2560 (control brain) · Photovoltaic solar panels × 6 (primary energy source) · Piezoelectric sensors × 30, 30 mm discs (kinetic energy harvest) · Solar charge controller (power conditioning) · Zener-diode voltage rectifier (spike protection) · 12 V rechargeable battery bank (energy storage) · IR infrared modules × 9 (occupancy detection) · SG90 servo motors × 2 (gate automation) · LDR optical sensor × 2 (adaptive lighting) · 16×2 LCD with I²C module × 2 (occupancy display) · C++ firmware with MQTT telemetry (software stack)

3.Results & Validation

Hard numbers build trust. Three KPIs at the top — then a 2-paragraph narrative of how you tested and what you found.



Example KPI values shown — replace numerator, denominator, and labels with your real metrics.

Voltage Accumulation: During a continuous 6-hour simulation, stored piezoelectric voltage rose from 142.3 mV in the first hour to a stable plateau of 328–348.5 mV by hours 3–6, sufficient to sustain microcontroller operation without any external supply.

Power Yield Growth: Harvested power grew from 329.8 μ W at hour one to a peak of 1,547.5 μ W (1.54 mW) by the sixth hour, confirming the system's capacity to accumulate progressively more energy as traffic activity increases.

All 10 functional test cases passed, including a gate servo response of 90° in 120 ms, slot-state registration in under 350 ms, IoT dashboard synchronisation with low MQTT latency, and successful voltage spike clamping by the Zener protection circuit.

4. Get Involved

Let's talk.

We are looking for industry partners, investors, and collaborators to take ECO Park LOT from validated prototype to full-scale deployment.

FOR INDUSTRY	FOR FUNDERS	FOR RESEARCHERS
Pilot With Us <i>Run a real-world pilot at your commercial parking facility. Full hardware setup and technical support provided by our team</i>	Fund the Next Phase <i>Help us scale from our EGP 7,315 prototype to a commercial deployment. Full Bill of Materials and financial projections available on request..</i>	Collaborate With Us <i>Co-author research, share vehicle traffic datasets, or extend the piezoelectric harvesting model into high-footfall pedestrian zones.</i>

CONTACT THE TEAM	
Team members	Ahmed Mostafa (240102998) · Ahmed Mohamed (240102989) · Mostafa Hisham (240102762) Mohamed Safwat (230100450) · Mohab Mohamed (230105667) · Mohamed Zakaria (240103011)
Supervisor	Dr. Mariam Ahmed Sameh · El Sewedy University of Technology
Project Page / Repo Qr Code for QMS	https://

Brief professional bios that build trust. Each member: name, role, one-line credential, what they did on this project.

Name	Role	Contribution & Background
Ahmed Mostafa	Team leader\ Hardware Lead	
Ahmed Mostafa	Energy Systems	
Mostafa Hisham	Software Lead	
Mohamed Safwat	Sensing & Control	
Mohab Mohamed	Testing & Validation	

Name	Role	Contribution & Background
Mohamed Zakaria	Feasibility & Design	-
Esam Ahmed	Feasibility & Design	

This project was developed at El Sewedy University of Technology (SUT) — Egypt's polytechnic university dedicated to applied engineering and industry collaboration. SUT students work on projects connected to real industrial partners across the energy, manufacturing, and technology sectors.